

P3↔P2 Instinct/Reasoning Separation Inheritance

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This work derives from and formalizes commitments across the CKS theory trilogy: The Instinct/Reasoning Separation Outside the Model (Li, April 2026), Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems (Li, April 2026), and Inter-Self Coordination via Shared Substrate / Full Aspect Integration (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the inheritance edge between Paper 3's instinct non-crossing commitment and exchange bounding, on the one side, and Paper 2's foundational instinct/reasoning separation, on the other — so that downstream work can locate these Paper 3 commitments precisely within the trilogy's prior-art structure.

Abstract

Paper 2 of the CKS theory trilogy establishes the instinct/reasoning separation as its foundational architectural commitment: every CKS-governed Self has an instinct layer (the LLM and its harness substrate, carrying organizational character and fast-pattern capability) and a reasoning layer (the substrate content — authored governance specifications and operational records — constituting the governed coordination space). The separation is architectural and absolute: instinct is not substrate content; reasoning is substrate content. Paper 3 extends this same separation to the inter-Self perimeter. The extension takes three operative forms: instinct non-crossing (the instinct layer cannot cross the organizational perimeter), exchange bounding (Full Aspect Integration operates only over reasoning-layer content), and FAI operating entirely within the reasoning layer (the shared substrate carries DNA-layer and action-layer content from multiple Selves; instinct-layer content never enters it). This note formalizes the inheritance edge. Paper 3's instinct non-crossing and exchange bounding are not new architectural principles: they are the direct operational application of Paper 2's instinct/reasoning separation at inter-Self scope. The separation is the same structural fact; its scope of application changes from governing how an LLM relates to its own Self's substrate to governing what can cross organizational boundaries.

1. Why this inheritance edge needs to be formalized

Paper 3's instinct non-crossing commitment and exchange bounding are each stated as architectural commitments in their own right in Paper 3 §5. Both can appear, at first reading, to be independent design choices — decisions Paper 3 introduces for reasons specific to inter-Self coordination, without necessary connection to Paper 2's foundational layer structure. This reading is incorrect, and its incorrectness has direct consequences for how any adversarial claim about AI governance content

exchange across organizational boundaries must be constructed.

The instinct/reasoning separation is Paper 2’s cross-claim spine: every subsequent claim in Paper 2 either establishes structures the separation requires, develops dynamics the separation enables, or specifies governance properties that keep the separation safe (Paper 2 §2.1). Paper 3 takes the Self — the coordinating unit Paper 2’s separation defines — as its participating unit. When Paper 3 asks what can cross the perimeter between two Selves, the answer is structurally determined by the separation: only what is in the reasoning layer can cross, because the reasoning layer is substrate content and substrate content is the medium through which inter-Self coordination operates (Paper 3 §4). Instinct-layer content is not substrate content; it therefore cannot enter the shared substrate.

Formalizing this inheritance edge names the derivation explicitly. Paper 3’s instinct non-crossing is not stipulated; it follows from the separation. Paper 3’s exchange bounding is not independently designed; it is the operational expression of “only reasoning layer can cross” applied at the inter-Self perimeter. The inheritance makes Paper 3’s commitments stronger, not weaker: they carry the full defensive prior-art coverage of Paper 2’s foundational claim, which itself extends Paper 1’s substrate/LLM division drawn at the governance boundary rather than the capability boundary (Paper 1 §2).

2. The source commitment: Paper 2’s instinct/reasoning separation

Paper 2 establishes, as its foundational architectural commitment, that every CKS-governed Self has two distinct layers that compose under unified human governance (Paper 2 §4).

The instinct layer is the LLM and its harness substrate — the organizational character and fast-pattern capability that is not directly authored as governance content. The instinct layer develops through instinct evolution (LLM upgrades and infrastructure changes). It shapes how the Self interprets and responds to its reasoning layer. Crucially, it is not substrate content: the LLM’s weights and behavioral dispositions cannot be directly governed through substrate content. They can only be influenced through the substrate’s orchestration of LLM inputs.

The reasoning layer is the substrate content — authored governance specifications in the DNA layer, operational records in the action layer, and all other structured substrate content. This is the governed coordination space. It is what human governance authority acts upon directly through the three rights (inspect, modify, override) that Paper 1’s human-governed commitment establishes.

The separation is architectural and absolute at Paper 2’s intra-Self scope. Instinct is not substrate content. Reasoning is substrate content. There is no third category. At intra-Self scope, the separation’s primary function is descriptive and governance-enabling: it defines why the LLM’s weights are not subject to direct governance through substrate content, and why the reasoning layer is the correct locus for human authority.

3. The extension: three operative expressions at inter-Self scope

Paper 3 takes the Self Paper 2 defines and asks how Selves coordinate across organizational perimeters. The medium of inter-Self coordination is a shared substrate — a temporarily constructed CKS substrate spanning more than one Self’s governance perimeter (Paper 3 §4). When a Self contributes to the

shared substrate, the contribution surfaces cells, DNA-layer content, and action-layer content from contributed aspects. The instinct/reasoning separation determines what is available to contribute: only reasoning-layer content is substrate content; therefore only reasoning-layer content can enter the shared substrate. Paper 3 makes this structural consequence explicit through three commitments:

Instinct non-crossing (Paper 3 §5). The instinct layer — LLM weights and instinct-layer content — cannot cross the inter-Self perimeter. Each Self’s organizational character is home-bound. This is the inter-Self scope application of the same structural fact established at intra-Self scope: instinct is not substrate content, and the shared substrate is a substrate. Nothing in the shared substrate’s construction changes the architectural category of instinct-layer material.

Exchange bounding (Paper 3 §5). Full Aspect Integration is bounded to reasoning-layer content: DNA-layer content (orchestration substrates, behavior substrates, schemas, rules) and action-layer content (recorded task instances, outputs, lived operational experience). The exchange boundary maps directly and precisely onto the instinct/reasoning separation. The bounding is not a restriction imposed on top of the separation; it is the separation expressed as a constraint on what can enter the shared substrate from any participating Self’s home substrate.

FAI at reasoning-layer scope (Paper 3 §7). FAI operates entirely in the reasoning layer: the shared substrate holds reasoning-layer content from multiple Selves, merge operations run over reasoning-layer content, and evolution-feed derived from FAI dissolution routes exclusively to reasoning-layer evolution mechanisms — DNA evolution and action-feedback evolution. Instinct evolution takes no FAI input by architectural commitment. The layer-routing rule (DNA-layer content feeds DNA evolution; action-layer content feeds action-feedback evolution; instinct evolution takes no FAI feed) is Paper 2’s three-evolution-mechanism structure operating with the instinct/reasoning boundary intact at inter-Self scope.

4. What is preserved and what is extended

Preserved. The architectural fact that instinct is not substrate content and reasoning is substrate content. This is Paper 2’s structural commitment, and it holds without modification at every scope the trilogy covers. At intra-Self scope, it defines what the LLM is relative to the substrate within one Self. At inter-Self scope, the same fact determines that the shared substrate — being a substrate — can carry only reasoning-layer content from participating Selves.

Extended. At Paper 2’s intra-Self scope, the instinct/reasoning separation governs the relationship between the LLM and its own Self’s substrate: how the LLM is positioned relative to what it operates over, and why the LLM’s weights are not subject to direct governance through substrate content. At Paper 3’s inter-Self scope, the same separation becomes a boundary condition on inter-organizational exchange: what can cross the organizational perimeter is exactly coextensive with what is in the reasoning layer. The separation’s function shifts from an intra-Self architectural description to an inter-Self exchange filter. This is a scope extension, not a new architectural principle.

Governance sovereignty as a consequence. Because the instinct layer is where each Self’s organizational character resides — the LLM and harness substrate that shapes how the Self interprets

all reasoning-layer content — exchange bounding that prevents instinct from crossing organizational perimeters protects each Self’s organizational character from external influence. A Self participating in FAI does not acquire a different organizational character as a result of the event. It may acquire new reasoning-layer content (DNA-layer or action-layer), which its existing instinct layer then processes under its existing governance configuration. The organizational character — the instinct layer — remains home-bound. This governance sovereignty property follows directly from the instinct/reasoning separation; it requires no additional commitment from Paper 3.

5. Operational test

A system instantiates the Paper 3 inheritance from Paper 2’s instinct/reasoning separation if and only if all of the following hold during any inter-Self coordination event:

1. Participating Selves contribute only reasoning-layer content (DNA-layer and action-layer substrate content) to the shared substrate; LLM weights and instinct-layer content are not contributed and do not appear in the shared substrate at any point during the event.
2. Any content that crosses an organizational perimeter — from a Self’s home substrate into the shared substrate, or from the shared substrate into a Self’s home substrate — is classifiable as DNA-layer or action-layer content under the taxonomy Paper 2 establishes, and not as instinct-layer content.
3. After FAI dissolution, evolution-feed at each participating Self routes DNA-layer content to DNA evolution and action-layer content to action-feedback evolution; no FAI-derived content is routed to instinct evolution at any participating Self.
4. Each participating Self’s instinct layer — its LLM and harness substrate — operates unchanged in architectural category by the event; the event may produce new reasoning-layer inputs for the instinct layer to process, but does not modify the instinct layer itself.
5. A system in which LLM weights, behavioral dispositions, or other instinct-layer content cross organizational perimeters during a coordination event does not instantiate this inheritance, regardless of whether it instantiates other aspects of the CKS inter-Self architecture.

6. Prior-art inheritance claim

Paper 3’s instinct non-crossing commitment and exchange bounding are direct extensions of Paper 2’s instinct/reasoning separation to the inter-organizational context. Any adversarial claim about what should be permitted to cross organizational perimeters in AI governance content exchange architectures must address this inheritance. What can cross is not an independent design decision in Paper 3: it is the operational consequence of an architectural separation Paper 2 establishes as foundational, which is itself a scope extension of Paper 1’s substrate/LLM division drawn at the governance boundary.

The inheritance chain is continuous: Paper 1’s substrate/LLM division at cell scope (Paper 1 §2) → Paper 2’s instinct/reasoning separation at Self scope (Paper 2 §4) → Paper 3’s instinct non-crossing and exchange bounding at inter-Self scope (Paper 3 §5). Each step extends the same structural fact to a larger coordination scope. A claim that instinct-layer content should cross organizational perimeters, or that the boundary between what can and cannot cross should be drawn differently, must engage Paper

2's foundational architectural commitment and demonstrate that a different drawing is compatible with the substrate-mediated coordination architecture Papers 1 and 2 establish — not merely that it is compatible with Paper 3 commitments considered in isolation.

Source papers

Li, W. (2026). *The Instinct/Reasoning Separation Outside the Model*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems*. April 2026. ORCID: 0009-0004-8065-3235.

Li, W. (2026). *Inter-Self Coordination via Shared Substrate / Full Aspect Integration*. April 2026. ORCID: 0009-0004-8065-3235.

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